Predicting When Collection Statistics Help Top-k Selection: From Retrieval to Attention and Expert Routing

Team ANLP Doomsday

Aviral Gupta Mohit Kumar Singh Anirudh Sankar Arihant Tripathy

{aviral.gupta, mohit.singh, anirudh.sankar, arihant.tripathy}@research.iiit.ac.in

IIIT Hyderabad, India https://github.com/avi1o1/ANLP-Doomsday

Abstract

Learned sparse retrieval, block-sparse attention and mixture-of-experts routing all select items by a top-k over inner products with learned non-negative features. Only lexical retrieval corrects this score, through the inverse document frequency (IDF), term-frequency saturation and length normalisation of BM25. Prior work disagrees on whether the correction helps learned features. We ask when it helps and whether that can be predicted without relevance labels. From one frozen multilingual encoder we build fourteen sparse representations and score each under all eight combinations of the three components on thirteen English and Indic collections. The full correction improves nDCG@10 in 90% of primary rows, yet a pre-registered predictor built on frequency diagnostics does no better than the majority sign. An exact Shapley decomposition explains the failure. Saturation, which uses no collection statistics, accounts for 61% of the gain. Its effect is set by the dispersion of the stored feature values, a relation that holds on held-out parameter settings in tests fixed in advance (Spearman 0.87 and 0.90). The contribution of IDF and length normalisation depends on relevance, which no index statistic captures. In exploratory analysis, agreement with a reference retriever predicts it on held-out corpora (R^2 of 0.52 to 0.66). A decision rule built on this proxy improves mean nDCG@10 by only 0.005 over always applying collection statistics. KV-cache selection and expert routing come next.

1 Introduction

A learned sparse retriever scores a query against a document by an inner product over a vocabulary of non-negative features and returns the top k (Formal et al., 2021b; Nguyen et al., 2023). Quest does the same for the pages of a key-value cache, scoring each page against the current query and attending only to the best ones (Tang et al., 2024). A mixture-of-experts router does it once more, scoring every expert and sending the token to the top k (Shazeer et al., 2017; Fedus et al., 2022). These methods grew up in different communities. They share one computation (Figure 1).

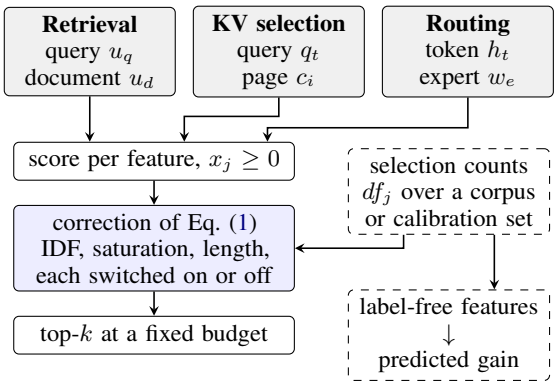


Figure 1: The computation shared by the three settings. Each scores an input against learned non-negative features and keeps the top k. The BM25 correction sits between the two steps. The selection counts it needs are also the input to the predictors of its gain.

Only retrieval learnt to correct the raw score in full. BM25 discounts features that occur throughout the collection, stops rewarding repeated evidence past a point and adjusts for document length (Spärck Jones, 1972; Robertson and Zaragoza, 2009). Each part was designed for words. Recent work applies BM25 to learned features and gets contradictory results. On one benchmark suite BM25 beats the inner product by 0.20 nDCG@10 on sparse-autoencoder features and loses by 0.11 on SPLADE-v3 (Clavié et al., 2026). The explanation offered is that BM25 helps while a vocabulary keeps a Zipfian frequency curve (Clavié et al., 2026; Han et al., 2026). If that account is right, statistics of the frequency curve should predict the sign of the gain before anyone runs the experiment. Our proposal made that prediction the central test.

This report tells what happened when we ran it. We built fourteen sparse representations from one frozen encoder and scored each under all eight

combinations of the three components on thirteen collections. The correction helped almost everywhere. The predictor failed. Working out why it failed turned into the main finding so far. Most of the gain comes from saturation, a component that reads no collection statistics at all. Its effect is set by the stored values, while the value of the two collection-level components depends on relevance. Section 7 follows the results in the order they came.

2 Problem Statement

The correction. For query and document vectors x_q, x_d with non-negative coordinates we score

$$S(q,d) = \sum_j x_{qj}\, w_j\, T(x_{dj}, \ell_d). \qquad (1)$$

The weight w_j is one or the smoothed inverse document frequency $\mathrm{idf}_j = \ln(1 + (N - df_j + 0.5)/(df_j + 0.5))$, where df_j of the N documents have feature j. The length factor ℓ_d is one or $1 - b + bL_d/\overline{L}$ with $L_d = \sum_j x_{dj}$. The transform $T(f, \ell)$ is f/ℓ or the saturated $(k_1+1)f/(f+k_1\ell)$. Each component is a switch, so a configuration is a bit string $c \in \{0,1\}^3$ in the order IDF, saturation, length. Configuration `000` is the plain inner product and `111` is BM25 (Robertson and Zaragoza, 2009). IDF and length normalisation read collection statistics. Saturation reads only the document.

The unit of observation. Collection frequency belongs to a corpus, so one row of the analysis pairs a representation with a collection at a support budget (the query and document feature caps), BM25 parameters (k_1, b) and a basis seed. Each row carries eight nDCG@10 values Q_c, one per configuration.

What is predicted. The pre-registered target is the margin $m = (Q_{111} - Q_{000})/\,\mathrm{sd}(Q_{000}, \ldots, Q_{111})$, a standardised effect on the same scale in every setting. Section 7 shows that m is poorly shaped for prediction, which led us to targets defined on the same eight values. The Shapley value ϕ_s of switch s is its marginal effect averaged over every order in which the switches can be turned on (Shapley, 1953). Writing Q_A for the configuration with the switches in A on,

$$\phi_s = \sum_{A \not\ni s} \frac{|A|!\,(2-|A|)!}{6} \big(Q_{A\cup\{s\}} - Q_A\big). \qquad (2)$$

The three values sum exactly to $Q_{111} - Q_{000}$. The collection gain $G = Q_{111} - Q_{010}$ is the value of IDF and length normalisation once saturation is applied. Its sign answers the practical question, whether to use collection statistics for a given representation and collection. A predictor maps label-free quantities to m, ϕ_s or G. We score it by sign accuracy against the majority sign and by held-out R^2 and Spearman correlation. For G we also measure the nDCG@10 reached when collection statistics are used only where the prediction is positive.

Transfer. Eq. (1) applies to any selection with a non-negative score and a selection count. For KV page i at decode step t, the feature value is the softplus of the Quest criticality $r_{t,i}$. Its frequency sf_i is the fraction of calibration queries for which the page enters the top-k among the pages that exist at that point. Its length is the norm of the page's concatenated key minima and maxima. For expert e at a token, the value is the router probability p_e, the frequency is the share of calibration tokens routed to e and the length is the norm of the router logits. The selected experts are combined with their original gate weights, so the correction changes only which experts run. In attention a row is a model, RULER task, context length and token budget, scored by RULER accuracy (Hsieh et al., 2024). In routing a row is a model and task, scored by negative token log-likelihood on WikiText and zero-shot MMLU accuracy. A predictor fitted in retrieval is applied there without refitting. The registered criterion asks for a sign accuracy of at least 70% and a mean absolute error of at most 1.5 times the corpus-held-out RMSE in retrieval. In retrieval, always predicting a gain already reaches 90% sign accuracy. So before any transfer run we add a third condition. The predictor must also beat the majority sign.

3 Motivation and Research Gap

The three selectors take the raw match score at face value. None asks how often a feature is selected across the collection, how evidence accumulates within an item or how large the item is. Lexical retrieval asks all three questions. Its answers arrived one at a time, as inverse document frequency (Spärck Jones, 1972), term-frequency saturation (Robertson and Walker, 1994) and pivoted length normalisation (Singhal et al., 1996), before BM25 combined them (Robertson and Zaragoza, 2009). Inside the transformer only fragments have reappeared. Loss-free balancing adds a per-expert bias

driven by recent load (Wang et al., 2024a), a frequency discount with no saturation or length term. Streaming inference exempts the first keys of the cache from eviction (Xiao et al., 2024).

Three papers have scored the same learned features with and without the correction, all in retrieval. BM25 wins on sparse-autoencoder features of a dense retriever (Clavié et al., 2026) and on SAE concept units (Park et al., 2025). It comes out level on visual words (Han et al., 2026). Two of these papers explain the pattern through the Zipfian shape of the vocabulary. None of the three varies the vocabulary systematically or asks which of the three components produces the gain. Their explanation has never been turned into a quantity computed before the experiment. Nor has it been tested where the same computation runs inside a model.

4 Contributions

1. **A controlled grid.** Fourteen sparse representations from one frozen encoder, scored under all eight configurations on thirteen collections, in 182 primary and 1,226 sensitivity rows. A second run on different hardware reproduces the primary rows it shares.
2. **A negative result with its cause.** The preregistered frequency predictor does no better than the majority sign. We show why it could not have worked.
3. **An attribution of the gain.** Exact Shapley values assign 61% of the gain to saturation and about a tenth to IDF. A mechanism for saturation, fixed in advance, passes its tests on held-out parameter settings.
4. **An exploratory label-free predictor** of the collection-level gain, built from agreement with a reference retriever.
5. **Transfer adapters** for KV-page selection and expert routing, with native-parity tests, ready for the frozen predictors.

5 Related Work

Term weighting. IDF began as a statistical account of term specificity (Spärck Jones, 1972). Saturation came from approximating the two-Poisson model of within-document frequency (Robertson and Walker, 1994). Length normalisation was pivoted so that long documents are neither favoured nor penalised (Singhal et al., 1996). BM25 combines the three (Robertson and Zaragoza, 2009). Every part was motivated by the statistics of words, which is why it is an open question whether any of them helps features that are not words.

Retrieval scorers. Dense retrievers score by the inner product of embeddings (Karpukhin et al., 2020). Our encoder is multilingual E5 (Wang et al., 2024b). Learned sparse retrieval keeps the inverted index and learns term weights, through contextual exact match (Gao et al., 2021) or expansion over the subword vocabulary in SPLADE (Formal et al., 2021b,a; Lassance et al., 2024). Two frameworks place these methods side by side (Lin and Ma, 2021; Nguyen et al., 2023). Learned weights also change the cost of query processing (Mackenzie et al., 2021), which is why we record posting visits and index size. Every one of these methods scores by inner product.

Learned concept vocabularies. Sparse autoencoders decompose dense activations into many non-negative features (Huben et al., 2024). Top-K SAEs fix the number active per input (Gao et al., 2025). CL-SR indexes SAE-decomposed embeddings as concept units weighted by IDF. Dropping that weighting costs 11 to 18 percent of its nDCG@10 on TREC-DL (Park et al., 2025), the one published ablation of the correction. SPLARE trains an SAE-based sparse retriever (Formal et al., 2026). SAE-SPLADE replaces the language-model head of SPLADE with an SAE (Zong et al., 2026). Both score by inner product. Quantised codes offer a second route to a discrete vocabulary, through product quantisation (Jégou et al., 2011), residual quantisation (Chen et al., 2010) and their GPU implementations (Johnson et al., 2021).

BM25 on learned features. Clavié et al. (2026) show that SAE features of a frozen dense retriever have roughly Zipfian collection statistics and can be scored by BM25 directly. In their Appendix C, BM25 reaches 0.582 nDCG@10 against 0.381 for the inner product on SAE latent terms. On SPLADE-v3 the order flips, 0.455 against 0.568. Han et al. (2026) find the two scorers within about two points of R@1 on image retrieval. They credit the parity to top-k filtering, which already suppresses what IDF would down-weight. When they raise the per-patch budget from 16 to 128, document frequencies converge and R@1 falls from 0.508 to 0.038.

Predicting effectiveness without labels. Query performance prediction estimates retrieval quality

from the query and the ranking alone (Carmel and Yom-Tov, 2010). One family of methods measures the divergence of the retrieved language model from the collection (Cronen-Townsend et al., 2002). Another reads the spread of the top retrieval scores (Shtok et al., 2012). Retrieval systems can also be ranked without judgements against pseudo-relevant documents drawn from pooled output (Soboroff et al., 2001). Our label-free predictor borrows both ideas for a new target, the difference between two scorers.

Attributing an effect to components. The Shapley value (Shapley, 1953), familiar from SHAP (Lundberg and Lee, 2017), is exact over a complete 2^3 factorial and needs no sampling.

Selection inside the transformer. Quest scores KV pages against page min-max key summaries (Tang et al., 2024). H2O keeps recent and heavy-hitter tokens (Zhang et al., 2023b). SnapKV keeps the positions an observation window attends to (Li et al., 2024). StreamingLLM keeps the first keys as attention sinks (Xiao et al., 2024). Expected Attention prunes pairs that future queries are predicted to ignore (Devoto et al., 2025). Massive activations single out a few positions by magnitude (Sun et al., 2024). None of these uses the rate at which an entry is selected across queries.

Expert routing. Sparse mixture-of-experts layers route each token to its top-k experts (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Balance comes from an auxiliary loss (Zoph et al., 2022) or from a load-driven bias (Wang et al., 2024a). DeepSeekMoE splits experts more finely and adds shared experts (Dai et al., 2024). Su et al. (2026) find a handful of experts, identified by activation magnitude, whose removal degrades the model badly. We use OLMoE (Muennighoff et al., 2025), which activates 8 of 64 experts per layer. To our knowledge no work treats retrieval scoring, KV selection and routing as one scoring problem.

Viability. Every piece of the study already existed, from cheap sparse codes to label-free quality proxies. What was missing was the controlled comparison, whose retrieval half is now complete.

6 Method

Representations. Holding the encoder fixed lets us vary only the vocabulary. All representations come from `multilingual-e5-base` (Wang et al., 2024b) at a pinned revision. Seven bases are fitted on the same 200,000 MS MARCO passages (Bajaj et al., 2016) and nowhere else. They are split-rectified identity, Gaussian projection (768), PCA (384), soft k-means over 4,096 centroids, product and residual quantisation (8×256) and a Top-K SAE with 8,192 features of which at most 32 are active. Pooled granularity applies each basis to the mean of the content token states. Token granularity applies it to each state before summing. This gives fourteen representations. A secondary SAE shares one dictionary across both granularities to separate aggregation from fitting. SPLADE has its own backbone, so it stays outside the grid as an external comparison.

Corpus statistics. Supports are capped before any statistic is computed, at 32 query and 128 document features in the primary budget. Document frequency, lengths and IDF come from the capped vectors of each target corpus. Retrieval is exact over the whole corpus. The primary grid uses $k_1 = 1.2$, $b = 0.75$ and seed zero. For a linear map C and diagonal weights D, weighted scoring $q^\top C^\top DCd$ preserves every ranking of the inner product when $C^\top DC = \alpha I$. Rectification, truncation, saturation and length normalisation are non-linear, so no representation here meets that condition.

Collections. The thirteen collections come from eleven corpora. MS MARCO dev, TREC-DL 2019 and TREC-DL 2020 share one set of documents. The others are six BEIR sets (Thakur et al., 2021), LIMIT (Weller et al., 2026) and the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a), each evaluated within its language. LIMIT is built so that the top-k sets its queries need exceed what a low-dimensional embedding can represent.

Runs. The primary family (Table 1) ran on a four-GPU cluster node for all thirteen collections. A single-GPU workstation reran it for the ten collections outside MS MARCO with the cluster run's fitted bases and pinned data. The workstation also ran every sensitivity family.

Significance. A paired bootstrap over queries recomputes numerator and denominator. Holm correction (Holm, 1979) runs within each pre-declared family, the all-on against all-off comparison and the three component main effects. The cluster run drew 2,000 samples, too few for any Holm p-value

Family	Rows
primary (14 representations $\times$ 13 collections)	182
support budget 16/64, 32/128, 64/256, native	462
$k_1 \in \{0.6, 1.2, 2.4\}$, $b \in \{0, 0.5, 0.75\}$	630
basis seeds 1 and 2	100
shuffled and uniform *df* controls	84
shared SAE dictionary	20
union on the ten non-MS MARCO collections	1,226

Table 1: Retrieval rows, each with eight scoring configurations. The primary count is the cluster run's. The other families and the union are the workstation run's. Families share their default cells, so the union is smaller than the sum. The MS MARCO budget rows are pending.

Collection	BM25	E5	`000`	`111`	best `111`
MS MARCO dev	.222	.425	.013	.130	.318
TREC-DL 19	.456	.704	.036	.255	.604
TREC-DL 20	.484	.697	.035	.251	.556
ArguAna	.411	.444	.021	.130	.371
FiQA	.236	.381	.013	.085	.266
NFCorpus	.307	.326	.064	.115	.285
SCIDOCS	.152	.172	.009	.044	.120
SciFact	.662	.694	.089	.197	.419
TREC-COVID	.582	.695	.134	.246	.508
LIMIT	.939	.015	.001	.004	.032
MIRACL hi	.357	.582	.011	.124	.328
MIRACL bn	.407	.700	.025	.216	.490
MIRACL te	.331	.751	.019	.259	.579

Table 2: nDCG@10 of text BM25, the dense encoder, the mean over the fourteen sparse representations under the inner product (`000`) and full correction (`111`) and the best representation under `111`. The best is the token-level SAE on every collection except LIMIT, where it is token k-means. All columns come from the cluster run.

over 182 comparisons to fall below 0.091. Significance is reported from the workstation run, which drew 10,000.

Diagnostics. No feature describing a row reads relevance labels. The six pre-registered frequency diagnostics are the active fraction, the Zipf slope of the feature-frequency curve, the Gini coefficient of activation mass, frequency and activation mass in the most frequent 1% of features and IDF variance. The primary ridge model uses the slope, IDF variance and frequent activation share. Index statistics describe the stored values, chiefly cv_x, the coefficient of variation of the stored document values. Ranking features come from the stored top-100 list $R_c(q)$ of every configuration c and query q. They measure how much each switch changes the top 10 and the shape of the top-100 scores (Shtok et al., 2012). The pseudo-quality $\tilde{Q}_c$ is the mean nDCG@10 of $R_c(q)$ when the top 10 of a reference retriever on the same query is taken as relevant. The references are text BM25, the dense encoder and the two fused by reciprocal rank. Putting $\tilde{Q}_c$ in place of Q_c in Eq. (2) and in G gives pseudo-label estimates $\tilde{\phi}_s$ and $\tilde{G}$ with no fitted parameter.

Validation. Predictors are ridge regressions with the scaler and penalty chosen inside training folds. Folds hold out a whole basis family (linear, clustered, quantised, learned) or a whole corpus, reported separately. A fit on primary rows is also scored on the rows at other k_1 and b, which share collections with it.

Transfer adapters. The attention adapter implements Quest-style criticality over sixteen-token pages with dense prefill and sparse decode. It counts how often each page enters the top-k over causal prefill positions and applies Eq. (1) with pages for documents. The routing adapter applies the correction to OLMoE router probabilities for selection and combines with the original gate weights. In both, the all-off configuration calls the native code path. Six tests confirm exact native parity, causal calibration, budget charging including sink pages and the gate mixture.

7 Progress

7.1 First, a check on the pipeline

We first checked that the pipeline gives the same answer twice. The workstation run reused the cluster run's bases and data on other hardware, so it tests the pipeline without being an independent replication. Over the 140 primary rows the two runs share, the median absolute difference in all-on nDCG@10 is 0.0007 (maximum 0.0077). The margin sign agrees in 136 of the 137 rows where it is defined. The differences come from fp16 rounding on different GPUs. Basis seeds move all-on nDCG@10 by a median standard deviation of 0.006 over 50 representation-collection groups.

7.2 The correction helps, because the inner product fails

The first thing the grid showed is that the plain inner product over capped sparse codes barely works (Table 2). Its mean nDCG@10 stays below 0.14 on every collection. Full correction raises that mean by a factor of 1.8 to 14. The margin is positive in 90% of the 179 primary rows where it is defined.

Over the 140 workstation rows the all-on against all-off comparison survives Holm correction in 99. The best representations become usable. The token-level SAE reaches 0.60 nDCG@10 on TREC-DL 2019, against 0.70 for the dense encoder it was built from. Token granularity is far stronger than pooled (mean all-on 0.219 against 0.097). The shared-dictionary rows show why. The pooled SAE reaches 0.095 with its own dictionary and 0.291 with the token-trained one, against 0.339 at token level. Most of the pooled deficit comes from fitting the dictionary on pooled vectors. On LIMIT, as its construction predicts, BM25 on words reaches 0.94 while the dense encoder and everything built from it stay near zero.

A closer look complicates the good news. Full correction is the best of the eight configurations in only 43 of the 182 primary rows. The most frequent winner, in 51 rows, is saturation with length normalisation and no IDF. The all-off configuration scores below 0.01 nDCG@10 in 42% of workstation rows. Almost any correction beats a scorer that broken, which is the first hint of trouble for the predictor.

7.3 The pre-registered predictor fails

The margin was meant to be predicted from the shape of the frequency curve. It could not be. The primary ridge model reaches held-out R^2 of 0.04 across basis families and 0.02 across corpora in the cluster run. In the workstation run, with the rows that respond to fewer than two switches excluded, the values are -0.11 and 0.14. Its sign accuracy equals the majority-sign baseline in every case (0.899 and 0.924). The six-diagnostic model reaches $R^2 = 0.30$ across corpora in the workstation run but -0.54 across basis families. Its sign accuracy is never more than 0.03 above the majority sign. By the criterion we registered, the frequency diagnostics carry no usable information about the margin.

Part of the failure lies in the target. The margin falls between 1.5 and 3.5 in 83% of workstation rows. It mostly records that the uncorrected scorer is broken, which leaves little variation for any diagnostic to explain. The rest of the failure, as the next subsections show, lies in what the correction is made of.

7.4 Where the gain comes from

To see what a predictor would have to capture, we split each row's gain among the three switches with Eq. (2). The answer was the same on both runs (Table 3). Saturation carries about 61% of the gain. IDF, the component the Zipfian account is about, carries about 10%. The main effects tell the same story, with means of $+0.065$ nDCG@10 for saturation, $+0.032$ for length and $+0.013$ for IDF, significant in 93, 79 and 56 of the 140 workstation rows. Once saturation is applied, collection statistics add 0.015 nDCG@10 on average in the workstation run and 0.021 in the cluster run. They help in about two thirds of rows.

Rows	IDF	Sat.	Len.	IDF + len.
cluster, 182 primary	.10	.62	.28	.38 (.36–.41)
workstation, 140 primary	.10	.61	.29	.39 (.36–.42)
budget 16/64	.16	.60	.24	.40
budget 64/256	.08	.57	.35	.43
native, 3 collections	$-.01$	.54	.47	.46

Table 3: Share of the all-on minus all-off gain assigned to each switch by Eq. (2). Intervals resample whole collections. The primary budget is 32/128. The other budget rows cover the ten workstation collections. Native support covers ArguAna, NFCorpus and SciFact.

The budget sweep shows the share of IDF shrinking as supports grow, from 16% at 16/64 to 8% at 64/256, while length rises from 24% to 35%. At native support IDF's share turns negative. Han et al. (2026) report the same trend for image features. IDF is not empty, though. Replacing the real document frequencies with shuffled ones on three collections turns the mean IDF Shapley value from $+0.011$ to -0.011. The real value is larger in 83% of 42 rows (one-sided Wilcoxon $p = 8.9 \times 10^{-7}$). Uniform frequencies remove IDF by construction and do not lower all-on nDCG@10 (0.157 against 0.147, $p = 0.15$). The frequencies carry information, but little of it reaches nDCG@10.

7.5 Why saturation can be predicted and the rest cannot

Table 4 suggested where to look. Saturation's contribution rises with cv_x, the spread of the stored values. The reason lies in the formula. With length normalisation off, saturation maps a stored value f to $g(f) = (k_1 + 1)f/(f + k_1)$. The ratio $g(f)/f$ falls as f grows, so saturation shrinks large values relative to small ones. If every stored value were equal, g would rescale all scores alike and change no ranking. The more unequal the values, the more the ranking moves. A smaller k_1 bends g harder. Pooled PQ and RQ give the extreme case. They

Basis	Gran.	111	cv_x	ϕ_{sat}	IDF + len.	$G > 0$
identity	pooled	.147	0.37	.043	.68	1.00
	token	.142	0.60	.060	.55	.77
Gaussian	pooled	.098	0.29	.020	.77	.85
	token	.088	0.55	.029	.64	.92
PCA	pooled	.222	0.75	.138	.24	.69
	token	.311	1.35	.155	.45	.46
k-means	pooled	.018	0.04	.000	.94	.31
	token	.185	1.71	.125	.11	.85
PQ	pooled	.012	0.00	.000	1.00	.62
	token	.144	1.44	.099	.09	.69
RQ	pooled	.056	0.00	.000	1.00	.92
	token	.292	2.10	.176	.18	.69
SAE	pooled	.127	0.18	.014	.66	.77
	token	.373	3.31	.193	.32	.46

Table 4: Per representation, mean all-on nDCG@10, mean saturation Shapley value, share of the gain from IDF and length and the fraction of collections where collection statistics help after saturation ($G > 0$), over the 13 primary collections. cv_x is the mean over the ten collections whose indexes were measured.

store equal values with equal lengths ($\mathrm{cv}_x = 0$). Their Shapley values for saturation and length are exactly zero in all 26 of their cluster rows and all 178 workstation rows. Outside these two codes, saturation helps in 91% of cluster primary rows. How much it moves the ranking is a property of the index. The move is almost always for the better.

IDF and length normalisation are different. They too move the ranking by an amount the index fixes. Whether the move helps depends on whether heavy documents were over-rewarded relative to their relevance and whether rare features separate relevant documents from the rest. Both questions involve relevance, which no statistic of the index records. This is why a predictor built on the frequency curve could not tell when collection statistics help. It also locates the decision that matters. For the two strongest representations, token-level SAE and PCA, collection statistics help after saturation on fewer than half the collections.

7.6 Testing the saturation account in advance

An account found by looking at the data needs a test the data did not shape. Before running any test we committed six hypotheses to the repository. All use the k_1 and b sweep, whose 630 rows the exploration had not touched. H2 to H6 are Holm-corrected together. H1 uses the default cell, which coincides with primary rows, so it checks consistency only. All six pass (Table 6), but they are not equally informative. H5 holds by construction, since the length factor is one at $b = 0$, so it checks the code. H2 and H6 test directions that follow from the form of Eq. (1). H3 and H4 carry the evidence. A model of saturation fitted on cv_x at $k_1 = 1.2$ still ranks the saturation gain at k_1 values it never saw, with Spearman 0.87 and 0.90.

7.7 Supplying relevance without labels

If the missing ingredient is relevance, a proxy for relevance should restore the prediction. Everything in this subsection came after the confirmatory tests were fixed and is exploratory. We scored every configuration against a reference retriever's top 10, treated as relevant. Table 7 shows the result. Index statistics predict saturation but fail for IDF and the collection gain. The frequency diagnostics predict none of the targets. Pseudo-labels predict all four, including G on held-out corpora (R^2 of 0.52 with the BM25 reference and 0.66 with the dense one). The mechanism is simple. Every target is a fixed linear combination of the eight Q_c, so a pseudo-quality that orders the configurations like the true one yields correlated targets. It does. The median within-row Spearman correlation between pseudo and true nDCG@10, over the eight configurations of 1,112 rows, is 0.98.

The practical test is a decision. A rule with no fitted parameter uses collection statistics only where $\tilde{G}$ is positive. On the primary rows it raises mean nDCG@10 from 0.1429 (always using them) to 0.1478, against 0.1279 for never using them and 0.1510 for an oracle. The gain of 0.0049 is 61% of the 0.0081 available, with a collection-resampled 95% interval of 0.0003 to 0.0101. The rule recovers 67% at other budgets and 65% at other parameters. At other seeds it recovers 22% and the interval reaches zero. The gains are small because always using collection statistics is already close to the oracle on most rows. We also chose the rule during the exploration.

The references carry biases of their own. The correction under study is BM25 applied to other features, so agreement with text BM25 may reward configurations for behaving like BM25. Lexical systems also contributed to the judgement pools of several collections. The dense reference errs the other way. The sparse codes come from the same encoder, so agreement with it partly measures how much of the embedding a configuration keeps. The prediction of G holds with either reference. With the dense one the rule recovers 45 to 75% of the headroom across the four groups of rows. Neither reference is free of bias, so the result

Week	Dates	Work	Owner
1	3–9 Oct	Resume the MS MARCO budget sweep (84 rows) after the cluster maintenance. Run SPLADE-v3 as the external comparison and the conceptual replication of the Appendix C reversal. Validate both adapters at scale, with native parity, calibration cost and memory.	Anirudh
		Commit the second pre-registration, the pseudo-label rule on the unseen MS MARCO budget rows, the frozen transfer predictors and the transfer thresholds.	Mohit
2	10–16 Oct	Setting B on Llama-3.2-1B-Instruct over all 13 RULER tasks with 500 examples, 4K and 8K contexts, budgets of 512 and 1,024 tokens, with H2O.	Anirudh
		Second confirmatory round in retrieval.	Mohit
3	17–23 Oct	Setting C on OLMoE-1B-7B, WikiText-2 negative log-likelihood and zero-shot MMLU, with the loss-free-balancing analogue.	Anirudh
		Write-up of the bilinear identity and its ranking conditions. Draft of the final report from the retrieval results.	Arihant
4	24–28 Oct	Transfer scoring with uncertainty for both predictors, per setting, against the majority sign.	Mohit
		Figures and tables from the saved artifacts. Viva slides.	Arihant, Aviral
5	29–31 Oct	Final report	Arihant
		Slides and presentation. Code archive.	Aviral, Anirudh

Table 5: Schedule to the final submission. If time runs short we cut, in order, GSM8K and expert preservation, the 8K context in Setting B, MMLU in Setting C (keeping WikiText) and the Appendix C replication.

	Hypothesis	Result
H1	IDF + length share at default in $[.36, .42]$	.393
H2	ϕ_{sat} at $k_1 = 0.6$ exceeds $k_1 = 2.4$	$.093 > .080$
H3	within-corpus $\rho(\text{cv}_x, \phi_{\text{sat}}) \geq 0.3$	.89, .89
H4	cv_x model ranks ϕ_{sat} at $\rho \geq 0.5$	.87, .90
H5	$\phi_{\text{len}} = 0$ exactly at $b = 0$	max 0.0
H6	ϕ_{len} at $b = 0.75$ exceeds $b = 0.5$	$.045 > .029$

Table 6: Confirmatory tests. H2 and H6 have Holm-corrected $p = 3.6 \times 10^{-5}$. H3 and H4 are tested at $k_1 = 0.6$ and 2.4. H4's model is fitted on primary rows at $k_1 = 1.2$. H5 covers 210 rows.

Target	Features	Family	Corpus	Params
ϕ_{sat}	pre-registered	$-.46$	.14	.02
	index	.47	.42	.50
	pseudo-labels	.56	.62	.76
ϕ_{len}	pre-registered	$-.30$	.14	.00
	index	$-.16$	.25	$-.02$
	pseudo-labels	.49	.70	.77
ϕ_{idf}	pre-registered	$-.01$	.00	.02
	index	.02	.02	.07
	pseudo-labels	.40	.21	.73
G	pre-registered	$-.07$	$-.02$	.16
	index	$-.05$	$-.09$	.15
	pseudo-labels	.38	.52	.62
	pseudo-labels, E5	.51	.66	.80

Table 7: Held-out R^2 of ridge regression fitted on 139 primary rows, holding out basis families or corpora, or predicting the 560 rows at other k_1 and b. Each tier adds features to the one above it. Pseudo-labels use text BM25 unless marked E5.

stays exploratory until it is tested on data fixed in advance.

8 Remaining Plan

The retrieval half of the project leaves two predictors to carry into the transformer. One is the pre-registered predictor, which goes to attention and routing as registered. The other is the pseudo-label predictor, which needs a label-free reference native to each setting. For KV pages that reference is exact attention mass. For routing it is the dense softmax mixture over all experts. Both come from the uncorrected model. Both predictors and the thresholds will be frozen before any corrected outcome is seen. Each result will be reported against the majority sign as well as the registered thresholds. Table 5 gives the schedule. Anirudh Sankar writes and runs the experiment code. Mohit Kumar Singh analyses the results and owns the pre-registrations. Arihant Tripathy writes the report and relates the results to the literature. Aviral Gupta, who proposed the idea, makes the presentations.

9 Limitations

One encoder underlies every representation, so basis and encoder effects are not separable. E5 saw MIRACL in training, so the Indic results say nothing about unseen languages. The sparse codes are far weaker than their encoder, so the attribution may not hold for strong retrievers such as SPLADE. The label-free predictor needs queries and a second retriever run. Routing is tested only at inference.

References

Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. 2016. MS MARCO: A human generated machine reading comprehension dataset. *arXiv preprint arXiv:1611.09268*.

David Carmel and Elad Yom-Tov. 2010. *Estimating the Query Difficulty for Information Retrieval*. Synthesis Lectures on Information Concepts, Retrieval, and Services. Morgan & Claypool.

Yongjian Chen, Tao Guan, and Cheng Wang. 2010. Approximate nearest neighbor search by residual vector quantization. *Sensors*, 10(12):11259–11273.

Benjamin Clavié, Sean Lee, Aamir Shakir, and Makoto P. Kato. 2026. Latent terms: Dense retrievers contain trivially extractable BM25-ready Zipfian vocabularies. *arXiv preprint arXiv:2605.29384*.

Steve Cronen-Townsend, Yun Zhou, and W. Bruce Croft. 2002. Predicting query performance. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 299–306. ACM.

Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li, Wangding Zeng, Xingkai Yu, Y. Wu, Zhenda Xie, Y. K. Li, Panpan Huang, Fuli Luo, Chong Ruan, Zhifang Sui, and Wenfeng Liang. 2024. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1280–1297, Bangkok, Thailand. Association for Computational Linguistics.

Alessio Devoto, Maximilian Jeblick, and Simon Jégou. 2025. Expected attention: KV cache compression by estimating attention from future queries distribution. *arXiv preprint arXiv:2510.00636*.

William Fedus, Barret Zoph, and Noam Shazeer. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39.

Thibault Formal, Carlos Lassance, Benjamin Piwowarski, and Stéphane Clinchant. 2021a. SPLADE v2: Sparse lexical and expansion model for information retrieval. *arXiv preprint arXiv:2109.10086*.

Thibault Formal, Maxime Louis, Hervé Déjean, and Stéphane Clinchant. 2026. Learning retrieval models with sparse autoencoders. In *International Conference on Learning Representations (ICLR)*.

Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. 2021b. SPLADE: Sparse lexical and expansion model for first stage ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2288–2292. ACM.

Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2025. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations (ICLR)*.

Luyu Gao, Zhuyun Dai, and Jamie Callan. 2021. COIL: Revisit exact lexical match in information retrieval with contextualized inverted list. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 3030–3042. Association for Computational Linguistics.

Donghoon Han, Eunhwan Park, and Seunghyeon Seo. 2026. Visual words meet BM25: Sparse autoencoder visual word scoring for image retrieval. *arXiv preprint arXiv:2603.05781*.

Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, and Boris Ginsburg. 2024. RULER: What's the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*.

Robert Huben, Hoagy Cunningham, Logan Smith, Aidan Ewart, and Lee Sharkey. 2024. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations (ICLR)*.

Hervé Jégou, Matthijs Douze, and Cordelia Schmid. 2011. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1):117–128.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2021. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online. Association for Computational Linguistics.

Carlos Lassance, Hervé Déjean, Thibault Formal, and Stéphane Clinchant. 2024. SPLADE-v3: New baselines for SPLADE. *arXiv preprint arXiv:2403.06789*.

Dmitry Lepikhin, HyoukJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. 2021. GShard: Scaling giant models with conditional computation and automatic sharding. In *International Conference on Learning Representations (ICLR)*.

Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.

Jimmy Lin and Xueguang Ma. 2021. A few brief notes on DeepImpact, COIL, and a conceptual framework for information retrieval techniques. *arXiv preprint arXiv:2106.14807*.

Scott M. Lundberg and Su-In Lee. 2017. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NIPS)*.

Joel Mackenzie, Andrew Trotman, and Jimmy Lin. 2021. Wacky weights in learned sparse representations and the revenge of score-at-a-time query evaluation. *arXiv preprint arXiv:2110.11540*.

Niklas Muennighoff, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Jacob Morrison, Sewon Min, Weijia Shi, Pete Walsh, Oyvind Tafjord, Nathan Lambert, Yuling Gu, Shane Arora, Akshita Bhagia, Dustin Schwenk, David Wadden, Alexander Wettig, Binyuan Hui, Tim Dettmers, Douwe Kiela, and 5 others. 2025. OLMoE: Open mixture-of-experts language models. In *International Conference on Learning Representations (ICLR)*.

Thong Nguyen, Sean MacAvaney, and Andrew Yates. 2023. A unified framework for learned sparse retrieval. In *European Conference on Information Retrieval (ECIR)*, pages 101–116. Springer.

Seongwan Park, Taeklim Kim, and Youngjoong Ko. 2025. Decoding dense embeddings: Sparse autoencoders for interpreting and discretizing dense retrieval. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 26468–26485, Suzhou, China. Association for Computational Linguistics.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.

Stephen E. Robertson and Steve Walker. 1994. Some simple effective approximations to the 2-Poisson model for probabilistic weighted retrieval. In *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '94)*, pages 232–241. Springer.

Lloyd S. Shapley. 1953. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games, Volume II*, number 28 in Annals of Mathematics Studies, pages 307–317. Princeton University Press.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and Jeff Dean. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*.

Anna Shtok, Oren Kurland, David Carmel, Fiana Raiber, and Gad Markovits. 2012. Predicting query performance by query-drift estimation. *ACM Transactions on Information Systems*, 30(2):1–35.

Amit Singhal, Chris Buckley, and Mandar Mitra. 1996. Pivoted document length normalization. In *Proceedings of the 19th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 21–29. ACM.

Ian Soboroff, Charles Nicholas, and Patrick Cahan. 2001. Ranking retrieval systems without relevance judgments. In *Proceedings of the 24th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 66–73. ACM.

Karen Spärck Jones. 1972. A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation*, 28(1):11–21.

Zunhai Su, Qingyuan Li, Hao Zhang, Weihao Ye, Qibo Xue, Yulei Qian, Yuchen Xie, Ngai Wong, and Kehong Yuan. 2026. Unveiling super experts in mixture-of-experts large language models. In *International Conference on Learning Representations (ICLR)*.

Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. 2024. Massive activations in large language models. In *Conference on Language Modeling (COLM)*.

Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. 2024. Quest: Query-aware sparsity for efficient long-context LLM inference. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235, pages 47901–47911. PMLR.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks (NeurIPS Datasets and Benchmarks)*.

Lean Wang, Huazuo Gao, Chenggang Zhao, Xu Sun, and Damai Dai. 2024a. Auxiliary-loss-free load balancing strategy for mixture-of-experts. *arXiv preprint arXiv:2408.15664*.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024b. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

Orion Weller, Michael Boratko, Iftekhar Naim, and Jinhyuk Lee. 2026. On the theoretical limitations of embedding-based retrieval. In *International Conference on Learning Representations (ICLR)*.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*.

Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2023a. MIRACL: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics*, 11:1114–1131.

Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023b. H2O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 34661–34710.

Yuxuan Zong, Mathias Vast, Basile Van Cooten, Laure Soulier, and Benjamin Piwowarski. 2026. From tokens to concepts: Leveraging SAE for SPLADE. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*, Melbourne, VIC, Australia. ACM.

Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam Shazeer, and William Fedus. 2022. ST-MoE: Designing stable and transferable sparse expert models. *arXiv preprint arXiv:2202.08906*.